PS200C: Quantitative Methods

In-class Activity 10: Sensitivity — Reading and Arguing

Name: ____________________

Setup. Recall the Darfur study from class: outcome *PeaceIndex* (0–1); treatment *DirectHarm*; adjustment for village and gender. The augmented regression table:

Treatment	Est.	SE	t	$R^2_{Y\sim D\mid\boldsymbol{X}}$	RV	$\text{RV}_{\alpha=0.05}$
Directly Harmed	0.097	0.023	4.18	2.2%	13.9%	7.6%

df = 783. Bound from *female* (1×): $R^2_{Y\sim Z\mid D,\boldsymbol{X}} \leq 12\%$, $R^2_{D\sim Z\mid\boldsymbol{X}} \leq 1\%$.

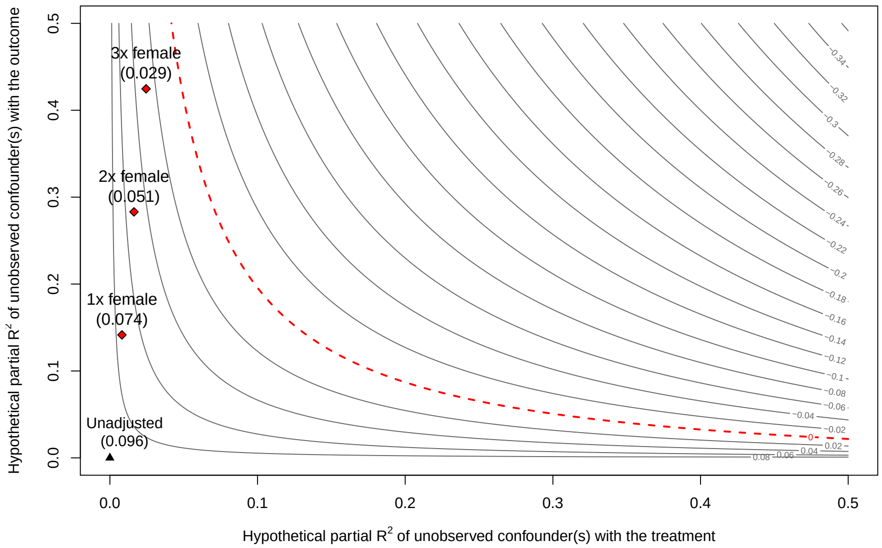
Question 1: Reading the table. Answer in one sentence each.

(a) What does RV = 13.9% tell us about the Darfur result?

(b) What does $\text{RV}_{\alpha=0.05} = 7.6\%$ tell us, and why is it smaller than the RV?

(c) What does $R^2_{Y\sim D\mid\boldsymbol{X}} = 2.2\%$ tell us about the role of confounding in this setting?

Question 2: Reading the contour. Below is the partial-R^2 sensitivity contour for the Darfur estimate. Each line is the adjusted point estimate at the corresponding pair $(R^2_{D\sim Z|\boldsymbol{X}},\, R^2_{Y\sim Z|D,\boldsymbol{X}})$. Diamond markers indicate hypothesized confounders $1\times$, $2\times$, and $3\times$ as strong as *female* in their associations with both D and Y.



(a) If unobserved confounding is twice as strongly related to the exposure (violence) and the outcome (peace index) as *female* is, then how would adjusting for such unobserved confounding change the estimated effect? Would it change your research conclusion?

(b) This contour shows what *would* happen *if* confounding were $k\times$ as strong as *female*. Why is it not sufficient to just show this (or any other benchmark)? What more needs to be argued for it to be meaningful?